

RAFAEL RANGEL DOS SANTOS FARINHA

AI Engineer

AI Engineer · RAG · Multi-agent · System prompt · Token economics · Evals · Scale

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github.com/Rafael-Rangel · rafael-rangel.github.io/portfolio

SUMMARY

AI and software engineer: I build LLM and RAG pipelines with embeddings, hybrid search, BM25, rerank, grounding, citations, and a knowledge base. I orchestrate multi-agent systems (planner/executor) with tool calling, function calling, system prompt, context engineering, prompt caching, token budget, token economics, model routing, and streaming, and I measure precision with evals, faithfulness, hallucination control, LLM-as-judge, and human-in-the-loop. I ship this in multi-tenant SaaS CRM and omnichannel (WhatsApp Cloud API, Instagram, Meta), with webhooks, REST APIs, queues, horizontal scale, load balancing, tenant isolation, ACL, RBAC, pgvector, and PostgreSQL — stack TypeScript, Python, React, Next.js, Node.js, NestJS, Docker, AWS, Azure, LLMOps, guardrails, PII, and prompt injection.

EXPERIENCE

Jul 2025 – Present

Software Engineering Director & Software Engineer

SaaS · Rio de Janeiro, Brazil

- Led AI engineering for CRM and support: multi-agent orchestration, tool calling, planner/executor, and human-in-the-loop.
- Shipped end-to-end RAG: ingestion, chunking, embeddings, hybrid search, rerank, grounding, citations, and answer faithfulness.
- Cut token cost with system prompts, context engineering, prompt caching, context compression, model routing, and token budgets.
- Raised agent precision with evals (tool-call accuracy, faithfulness, prompt regression) and retrieval-failure fallbacks.
- Scaled inference and queues per tenant: context isolation, knowledge-base ACLs, streaming, retries, and LLMOps observability.

Jul 2024 – Jul 2025

Front-end Developer

Web · Rio de Janeiro, Brazil

- Built e-commerce with 5,000+ products, catalog management, Bling integration, POS in 3 stores, and factory inventory fully synchronized.
- Built a delivery system with payment gateway, Pagar.me split, freight calculation, and Correios integration.

FEATURED PROJECTS

KoruVision CRM — github.com/Rafael-Rangel/koruvision-landing

CRM SaaS with AI agents, Workflows, RAG, and omnichannel support (WhatsApp/Instagram).

Rubi 2.0 / Conversation AI — github.com/Rafael-Rangel/rubi-teste

Specialized agents, knowledge base, precision evals, and human handoff.

Conversational English for Devs — github.com/Rafael-Rangel/ingles-conversacional-devs

Tutor with embeddings, controlled context window, and system/user prompt practice.

EDUCATION

Analysis and Systems Development — FIAP. Completed in 2025 (2-year degree).

CERTIFICATES

- Professional Qualification in Technology Strategy and Innovation with AI and IoT (2025)
- Professional Qualification in Mobile Application Development (2025)
- Professional Qualification in Web 2.0 Analysis and Design (2024)
- Professional Qualification in Systems Analysis and Web Prototyping (2024)
- Professional Qualification in Systems Analysis and Web Prototype Design (2024)
- Professional Qualification in Web 2.0 Development and Design (2024)

TECHNICAL SKILLS

- **AI:** RAG, embeddings, hybrid search, rerank, multi-agent, tool calling, function calling, system prompt, context engineering, evals

- **Tokens & quality:** token budget, prompt caching, model routing, streaming, faithfulness, hallucination control, LLM-as-judge
- **Scale:** queues, async processing, pgvector, PostgreSQL, tenant isolation, ACL, LLMOps observability
- **Stack & cloud:** Python, TypeScript, OpenAI, Node.js, NestJS, AWS, Azure, Docker, guardrails, PII, prompt injection